Sahaj Bhadja

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Summary

Applications Software Engineer with 2+ years developing, testing, and supporting business applications using **TypeScript/Node.js**, **React**, **SQL**, and **REST APIs** across workflow automation, integrations, and production troubleshooting.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Architected a multi-tenant business application on **Azure SQL** and Blob Storage with row-level isolation and normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built and maintained FastAPI integrations across PostgreSQL, **SQL Server**, and cloud storage, processing 5K+ monthly jobs with validation, reconciliation, error handling, and CI/CD releases sustaining 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Developed **React** interfaces and Flask services for uploads, grading history, and feedback, reducing turnaround 60% while supporting 150+ instructors and 10K+ daily requests reliably in production on AWS.
- Implemented concurrent batch processing and secure PDF, DOCX, and ZIP ingestion through **REST APIs** and AWS RDS, executing 1K+ files per batch with fault-tolerant recovery and usage tracking.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- **Troubleshot** and optimized large-scale API integrations using object-oriented patterns, AWS Glue, and S3, improving response times 30%, pipeline efficiency 68%, and maintainability while reducing technical debt 20%.
- Automated nightly builds, installations, **unit tests, integration tests**, and reports through Jenkins across Windows, Linux, and macOS, collaborating with Microsoft and GitHub stakeholders throughout Agile release cycles.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight **Node.js** microservices for authentication, resources, customer records, billing, leads, permissions, and notifications, synchronizing 2M+ records through REST APIs and WebSocket updates across six operational workflows.
- Automated Docker builds, unit, integration, and end-to-end tests, and staged deployments through **Jenkins CI/CD**, achieving 95%+ validation while using Sentry, Azure Monitor, and Kibana for production troubleshooting.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Translated **client requirements** into an end-to-end application architecture, mapped 12+ integration points, and presented technical solutions before implementation, reducing design errors 25% through collaborative cross-functional planning sessions.
- Built Flask APIs and Firebase services for cross-platform booking workflows, integrating pricing, coupons, payments, webhooks, and real-time status updates across 50K+ records with sub-200-millisecond responses across mobile and web clients.

Skills

Languages & Web: TypeScript, JavaScript, Java, Python, SQL, HTML, CSS, React, Node.js, Flask, Spring Boot
Applications: REST APIs, WebSockets, Webhooks, System Integrations, Microservices, Workflow Automation, Technical Troubleshooting, Application Support, Requirements Analysis, Cross-functional Collaboration
Data & Cloud: PostgreSQL, SQL Server, Azure SQL, SQLite, MySQL, MongoDB, Firebase, AWS RDS, Microsoft Azure
Quality & Delivery: Unit Testing, Integration Testing, Regression Testing, PyTest, JUnit 5, Jest, Selenium, CI/CD, Jenkins, GitHub Actions, Git, Docker, Agile/Scrum, Technical Documentation, UAT

Projects

OpsRelay | *TypeScript, Node.js, REST APIs, SQLite, SQL, JavaScript*

- Built a **TypeScript/Node.js** operations workflow platform with SQL state transitions, quality holds, audit trails, and idempotent integrations; processed 541,909 public transactions into 25,900 work items across 24 passing tests.

Chat2SQL Code | *Java, TypeScript, Node.js, SQL, PostgreSQL, Docker*

- Built a Java and TypeScript database workspace supporting 10+ relational and NoSQL systems, accelerating query authoring, schema exploration, and data management through reusable services and a unified cross-platform interface.

HMIS Live | *TypeScript, MERN, ABDM APIs, Sentry, Twilio*

- Built a TypeScript healthcare operations application integrating ABDM APIs, appointment workflows, Twilio notifications, and Sentry monitoring, supporting 100+ daily appointments while reducing administrative workload 30% for clinical teams.